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IFC Monitor – An IFC schema extension for modeling structural health monitoring systems

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ABSTRACT

The pervasive emergence of sensing technologies for structural health monitoring (SHM) and the digitalization ubiquitous in engineering ("Industry 4.0") pose increasing demands on information modeling concepts in civil engineering. While in building information modeling (BIM) conventional building information (such as geometry, material, or cost) can precisely be described using current modeling standards, information about SHM systems, referred to as "monitoring-related information", cannot be fully described on a well-defined, formal basis. In this paper, a BIM-based approach towards describing monitoring-related information is proposed, using the Industry Foundation Classes (IFC), an open BIM standard facilitating the interoperability of BIM models, as a formal basis. First, possibilities and constraints of describing monitoring-related information with the IFC schema are discussed. Then, information necessary to describe SHM systems is integrated into a semantic model serving as a technology-independent metamodel. Next, the IFC schema is extended to enable BIM-based descriptions of SHM systems in compliance with IFC modeling capabilities, which is referred to as "IFC Monitor" schema. The IFC Monitor schema is verified with test software used in the official IFC certification program. For validation, a prototype SHM system is formally described using the IFC Monitor schema. The validation aims at checking if the IFC Monitor schema is capable of precisely describing monitoring-related information. As will be shown in this paper, the description of the prototype SHM system meets the requirements of a well-defined IFC model as specified in the official IFC certification program. As a result, the IFC Monitor schema proposed in this study advances BIM-based descriptions of SHM systems in association with structural systems being monitored on a well-defined, formal basis.

1. Introduction

Structural health monitoring (SHM) has become an integral part of smart structures. The term "smart structures" denotes buildings (or infrastructure) equipped with sensor technologies to monitor and to control structures in real-time, using intelligent, networked processes [1]. Smart structures, gaining increasing popularity within the Industry 4.0 trend, are considered one unit comprising both the structure and the SHM system installed on the structure [2]. Due to decreasing cost of computer, sensing and communication technologies, SHM systems can be cost-efficiently designed and installed to monitor the behavior and to evaluate the condition of structures [3]. In the past decades, SHM has been successfully applied to large-scale engineering structures, such as bridges, wind turbines, and tunnels [4–7]. The performance and the availability of SHM systems have been substantially improved in recent years through applying artificial intelligence techniques, autonomous software, biologically-inspired approaches, and agent-oriented concepts [8].

With the ubiquitous availability of sensor-based systems for SHM, several standards, modeling languages, and ontologies have been developed to describe and to document sensor data. For example, with the Sensor Model Language (SensorML), sensors and processes associated with sensor measurements can be described [9]. With the Semantic Sensor Network (SSN) ontology, a framework is provided for modeling sensors and sensor networks. In the SSN ontology, sensor-related information is described in terms of capabilities, measurement processes, observations, and deployments [10]. Furthermore, formalized within the IEEE 1451 standards family, the description of network-independent communication interfaces for connecting sensors (and other transducers) to microprocessors, instrumentation systems, and control/field networks is supported [11].

While sensor data can precisely be modeled, stored, and exchanged, for example using database development methodologies [12], the description of monitoring-related information is still in its infancy. The term "monitoring-related information" refers to information required to describe SHM systems, such as information about the semantic

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composition of SHM systems, about embedded algorithms of sensor nodes deployed, or about the dynamics of SHM systems [13–15]. With a formal description of monitoring-related information, tracking of changes in SHM systems or substitution of components of SHM systems can be simplified. However, a description of SHM system components in the context of the structural system being monitored, which has not yet received adequate attention, cannot be fully achieved with current methods [16,17].

In recent years, building information modeling (BIM) has become very popular, providing means widely used in the architecture, engineering, and construction industry to describe conventional information about structural systems. BIM comprises tools, processes, and technologies for documenting and exchanging 3D digital building and civil infrastructure models [18]. Therefore, BIM is typically based on 3D geometric models supplemented by semantic information of structural components visualized in the geometric models. BIM provides a methodology for interdisciplinary metamodeling to describe different kind of information from different disciplines, which renders BIM an attractive formal basis to describe monitoring-related information. Through digital model-based BIM workflows, model changes can be evaluated in real time. By linking elements of SHM systems (e.g. sensors) with external sources (e.g. sensor data stored in databases), effective data management can be achieved, which can be used, for example, to visualize states of building elements [19].

Usually, BIM processes are classified according to whether the processes, being spatially distributed, are attuned to the same software ("closed BIM") or to cross-software applications ("open BIM"). An advantage of closed BIM processes is the coordination of models from all planners through the same file format, which is typically a proprietary file format of a specific software manufacturer. However, disadvantages of closed BIM processes are the restrictions and the limited flexibility caused by these restrictions. For example, certain software tools may not be available within a software package a closed BIM process is attuned to; in this case, the building information models are transferred from one application to another application that supports the software tools needed to work on the building information models. However, inter-application transfer of building information models may cause a loss of semantic information.

In contrast to closed BIM processes, the Industry Foundation Classes (IFC) support platform-independent, open BIM processes. The IFC are standardized under ISO 16739:2013 in its current version "IFC 4 – Addendum 2" to describe and to exchange building information models [20]. With the IFC, a standard for comprehensive semantic models has been established describing building information in all phases of a life cycle of a building. The IFC adopt an object-oriented approach, in which building information is assembled as a set of objects, each object containing attributes describing the object. In addition to objects, the IFC schema provides a predefined set of types, rules, and functions used to capture information relevant to the domain of interest, e.g. architecture or structural engineering.

Because the IFC have primarily been designed to describe buildings, extensions of the IFC schema are developed to integrate information about infrastructure, such as bridges, roads, railways, and tunnels [21–25]. With SHM systems being considered a specific type of infrastructure, it is appropriate to use the IFC schema as a basis to describe SHM systems. Hence, SHM systems can be described in a single model along with the structures being monitored. While single sensors can be described with the IFC, the IFC schema does not support a full description of monitoring-related information. Rio et al. [26] have evaluated the IFC schema to describe sensor-related information. As a result, Rio et al. have concluded that the absence of modeling capabilities of the IFC schema restricts the description of SHM systems. Thus, there is a need to develop well-defined standards to describe monitoring-related information. A similar conclusion is drawn by Del Grosso et al. [27] who integrate BIM processes with SHM systems to improve infrastructure management processes. Another approach adopted by Huston et al. [28] highlights the development of modeling capabilities for SHM systems in the context of BIM as a major research challenge.

To describe monitoring-related information with the IFC schema, Delgado et al. [29] have proposed using generic IFC constructs, such as proxy elements and user-defined property sets. However, using proxy elements and user-defined property sets, being unspecific and non-standardized, may cause a loss of semantic information. To describe monitoring-related information without losing semantic information, this study presents an IFC schema extension, referred to as "IFC Monitor". The IFC Monitor extension aims at facilitating the documentation of SHM systems and tracking of changes that SHM systems undergo over the operational life. As a further advantage, as compared to approaches combining different ontologies and metamodels with self-defined semantic models [30], all information needed for a well-defined description of SHM systems is contained in a single, IFC-based building information model. Therewith, in a typical exchange scenario, planners of SHM systems have the opportunity to document designs of SHM systems in a standardized manner.

This paper is organized as follows. First, SHM-related modeling capabilities of the IFC schema are summarized in a concise literature review. Then, a semantic model is proposed, serving as a technology-independent metamodel to formally describe SHM systems, whose focus is emphasized (i) on semantic compositions of SHM systems, (ii) on network topologies, and (iii) on semantic relationships between components of the SHM system and components of the structural system being monitored. Based on the semantic model, the IFC Monitor extension is designed to enable BIM-based descriptions of SHM systems in compliance with the IFC schema. For verification, the extended IFC schema is checked with test software that is used in the official IFC certification program [31,32]. Next, serving as a test bed for the BIM-based modeling approach, a prototype SHM system is formally described using the proposed IFC Monitor extension, leading to an IFC model of the prototype SHM system. Finally, the IFC model is validated according to whether it meets the requirements of a well-defined IFC model.

2. SHM-related modeling capabilities of the current IFC schema

The IFC schema, to a limited extent, provides SHM-related modeling capabilities, which are summarized in this section. Upon a review of general IFC modeling capabilities, it is evaluated how information beyond the scope of the current IFC schema can be described. The final subsection focuses on the description of monitoring-related information using the current IFC schema.

The IFC schema follows an object-oriented approach, containing several classes that serve as "templates", hereinafter referred to as "entities", to describe building information. The semantic meaning and the implementation of the entities are generalized using inheritance mechanisms. Describing the most abstract fundamental entity in the IFC schema, the *IfcRoot* entity is used as a basis for other entities. The *IfcRoot* entity, illustrated in Fig. 1, provides capabilities for (i) identifying entities by global unique identifiers, (ii) defining ownership and change information of entities, and (iii) attributing names and descriptions of entities.

In the IFC schema, entities are distinguished according to whether the entities represent objects, properties, or relationships: *Objects* represent any "thing" semantically described; *properties* cover all characteristics that may be assigned to objects; and *relationships* among objects are described as "objectified relationships". An objectified relationship is a formal construct that describes a relationship between objects by a relationship entity that points to all objects related with each other, i.e. there are no explicit attributions that describe relationships between objects. As a consequence, the semantic meaning of objectified relationships is implicitly deduced by the types of relationship entities, e.g. a relationship describing the decomposition of an object into single parts of the object, as exemplarily illustrated in Fig. 1

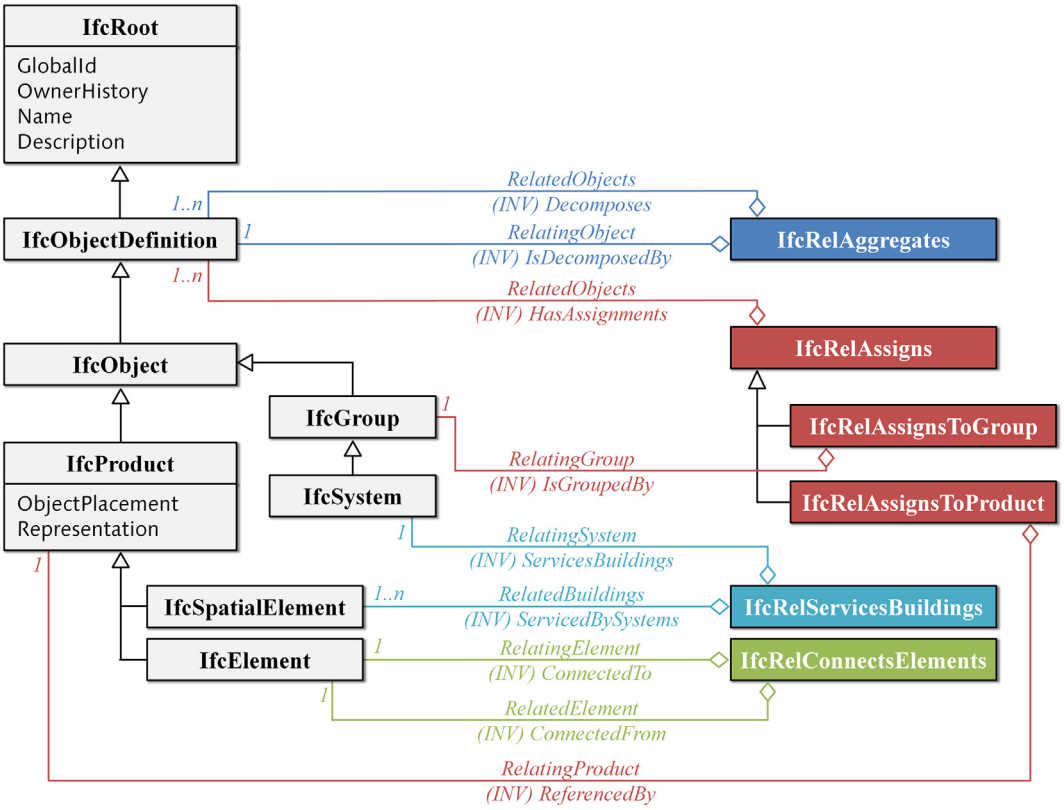


Fig. 1. Object entities (gray) and relationship entities (colored) of the IFC schema relevant to monitoring information modeling. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

by the relation “is decomposed by” and by the relation “decomposes” in opposite direction.

Because navigation in object models without explicit attributions is time-consuming and causes considerable computational cost, so called “inverse attributes”, labeled “(INV)” in Fig. 1, are introduced into the IFC schema to compensate for the absence of explicit attributions. Inverse attributes describe the opposite relation between an object on the one side and a relationship entity on the other side. Inverse attributes can automatically be evaluated by software tools based on the formal definitions taken from the IFC schema.

The IFC schema contains different types of object entities, such as structural components, construction processes, actors, groups, and resources. In the following paragraphs, object entities and relationship entities relevant to the IFC Monitor extension proposed in this study are discussed.

In the IFC schema, object entities are subtypes of the abstract entity *IfcObject*. As shown in Fig. 1, object entities can be decomposed into “smaller” components using the relationship entity *IfcRelAggregates*. Objects (*IfcObject*) that appear in a spatial or geometric context are described by abstract entities of type *IfcProduct*. Geometric representations and locations may be assigned to *IfcProduct* entities. As a more specific subtype of *IfcProduct*, architecture, engineering, and construction (AEC) products are described by the abstract entity *IfcElement*. *IfcElement* entities can be connected to other IFC elements, e.g. to describe connections between elements, using the objectified relationship entity *IfcRelConnectsElements*. Specific subtypes of *IfcElement* entities are used to further distinguish the semantic meaning of an AEC product, including object entities to describe building elements (e.g. walls, slabs, or beams), furnishing elements, or distribution system elements. As opposed to AEC products, elements describing spatial structures, such as sites, buildings, stories or spatial zones, are defined by subtypes of the abstract entity *IfcSpatialElement*.

Semantic references from objects (i.e. subtypes of *IfcObject*) to products (i.e. subtypes of *IfcProduct*) are typically modeled using relationship entities of type *IfcRelAssignsToProduct*. Using the entity *IfcGroup*, distinct collections of objects can be aggregated to groups under topological, non-geometrical grouping aspects. Relationships between groups and objects are described with the relationship entity *IfcRelAssignsToGroup*. As a subtype of *IfcGroup*, the entity *IfcSystem* describes a specialization of a group, with the parts of the group assembled for a common purpose to provide a certain service. Finally, the relationship entity *IfcRelServicesBuildings* can be used to describe the semantic relationship between a system and spatial elements, such as spatial zones, stories, or buildings, that are connected to the system.

The IFC schema is designed to support the description of building information in different domains, such as architecture, electricity, or building control. The following subsection illustrates possibilities provided by the IFC to describe information that is out of scope of the IFC schema or that exhibits a level of detail not covered in the IFC schema, which is, in the following, referred to as “supplementary information”. Thereupon, capabilities provided by the IFC are presented to describe monitoring-related information.

2.1. IFC-based description of supplementary information

The IFC provide three constructs for describing supplementary information in a generic way: Proxy elements, object types, and property sets. In the following paragraphs, the IFC constructs are analyzed with respect to describing supplementary information, both in general and with regard to SHM systems.

Proxy elements are either placeholders without predefined semantics or containers used to wrap object entities. Using proxy elements, representing “non-standard” IFC objects, for describing supplementary information may result in a loss of semantic information. Thus, proxy elements are considered not suitable for a semantically well-defined description of non-standard IFC objects.

Object types enable semantic distinctions of object entities without the need for additional subtypes. Therefore, attributes that further differentiate object types are introduced into the IFC schema by several object entities. Depending on the object entity, sets of predefined enumerators are provided to specify the object types. For example, a sensor in the IFC schema described by the entity *IfcSensor* can be associated with the predefined enumerator *WINDSENSOR*. However, if describing complex SHM systems it is likely that enumerators for precisely denoting the appropriate sensor type (or object type) are not available from the IFC schema; in this case, a user-defined key may be used for denoting the object type. Since any alphanumeric strings may be assigned to object types, conventions are required to render the information computer-interpretable. In conclusion, using user-defined object types for describing supplementary information can be ambiguous and error-prone.

Property sets, representing a generic approach based on key-value pairs, allow attaching supplementary information to IFC entities. For property sets, predefined keys (and corresponding data types for the values) are defined in the IFC schema. Similar to object types, user-defined keys (and value types) may be used for property sets to describe supplementary information. However, user-defined IFC property sets, similar to object types, may be ambiguous and error-prone, since any alphanumeric string can be assigned to the key of the key-value pair (and the value can be of any type).

To compensate for the ambiguity of user-defined object types and user-defined property sets, data dictionaries, such as the buildingSMART Data Dictionary (bSDD) [33], and information delivery manuals have been proposed [34]. Data dictionaries and information delivery manuals are designed to advance the description of objects, properties, and processes, independent from natural languages. However, data dictionaries and information delivery manuals are not available in every BIM software tool used in the modeling process. Furthermore, data dictionaries and information delivery manuals focus on certain domains of interest. Thus, a well-defined unambiguous description of information is not possible, if the information is out of scope of the domains of interest.

In summary, supplementary information can partially be described with proxy elements, with user-defined object types, and with user-defined property sets. However, partial description of supplementary information may cause a loss of semantic information. Furthermore, all constructs mentioned above are not suitable for automated data processing due to their ambiguous nature (i.e. no predefined semantics are inherent to proxy elements and no conventions about user-defined object types or user-defined property sets are defined). Although the degree of ambiguity may be reduced using data dictionaries and information delivery manuals, ambiguities cannot be completely resolved. As will be shown in the following subsection, some aspects of monitoring-related information can be described using the constructs provided by the current IFC version, which is, however, not sufficient for correct and precise descriptions of SHM systems.

2.2. IFC-based description of monitoring-related information

Because the focus of the current IFC version is put on the description of buildings (including building automation systems) and infrastructure, a comprehensive description of monitoring-related information cannot be achieved using the modeling capabilities offered by IFC. In the following paragraphs, the modeling capabilities predefined in IFC that are relevant for describing monitoring-related information are further elucidated, including, for example, description of sensors, communication units, and power units.

A sensor description is provided by the IFC entity *IfcSensor*. A sensor is classified by assigning an object type to the sensor. In total, 24 predefined object types for sensors are available in IFC, such as temperature sensors or humidity sensors [35]. However, sensor types frequently used for SHM, such as accelerometers or strain gauges, are not predefined in IFC. As mentioned earlier, user-defined object types can be used for denoting non-predefined object types. However, due to the ambiguity of user-defined object types, a loss of semantic information may occur, preventing automated interpretation of information.

Sensor properties, such as information about sensor manufacturers or information about energy consumption, can be described in IFC using predefined property sets. Some predefined property sets are applicable to all sensor types, while other property sets that describe more specific sensor information are applicable only to specific sensor types. For example, temperature values recorded to indicate fire solely are applicable in association with a predefined “fire” sensor type. To overcome this restriction, user-defined property sets, similar to user-defined object types, may be used for describing information about sensor properties not predefined in the IFC. However, due to the ambiguity of user-defined keys, automated interpretation and processing of monitoring-related information may be hampered if applying user-defined property sets.

In addition to sensors, the IFC provide cross-domain entities that can be re-used for describing SHM systems, such as object entities to describe communication units or power supply units. For example, communication units can be described by *IfcCommunicationsAppliance* entities. By assigning an object type to communication units, the semantic distinctions of communication units can be further specified using predefined enumerators (e.g. *COMPUTER*, *ANTENNA*, *GATEWAY*, *ROUTER*). Cable-based power connections between elements can be described by *IfcCableSegment* entities (object type: *CABLESEGMENT*), internal power supply units can be described by *IfcElectricFlowStorageDevice* entities (object type: *BATTERY*), and external power sources can be described by *IfcOutlet* entities (object type: *POWEROUTLET*).

3. A semantic model for SHM systems

In this section, a semantic model for SHM systems is proposed, facilitating the IFC-based description of monitoring-related information on a formal basis. The semantic model, representing a metamodel for modeling SHM systems, is developed based on a three-step process that includes (i) a review of SHM regulations and guidelines, (ii) a review of existing sensor models and modeling languages, and (iii) a definition and classification of specific monitoring-related information relevant to SHM systems. For details on the design process of the semantic model, the interested reader is referred to previous work of the authors presented in [13,16,36].

As can be seen from Fig. 2, a BIM model in the context of SHM describes a SHM system and the structural system that is monitored by the SHM system. A SHM system is composed of a computer system and one or more sensor networks. The computer system, which includes processes and resources, is used to manage and to analyze sensor data. Sensor networks, essentially, are composed of collections of nodes (described by the abstract type *Node*), each node accomplishing distinct tasks in a sensor network. Furthermore, sensor networks may assume semantic references to the structural system, e.g. to spatial elements of the structural system. Sensor networks are characterized by network topologies that describe node connections and communication paths. In Fig. 3, different types of network topologies supported by the semantic model are shown, such as line topology or star topology.

In Fig. 2, two node types are introduced, sensor nodes and base stations. Sensor nodes are devices that collect and process sensor data, control actuators, and communicate with other nodes. Base stations are introduced to enable wireless communication between sensor nodes and computer systems on site. Further means of communication, such as cloud-based communication gateways, are out of the scope of this paper. Independent from the node type, each node requires a power unit, a communication unit, and a node specification that defines location and geometry of the node. In addition, semantic references from nodes to components of the structural system are defined. Furthermore,

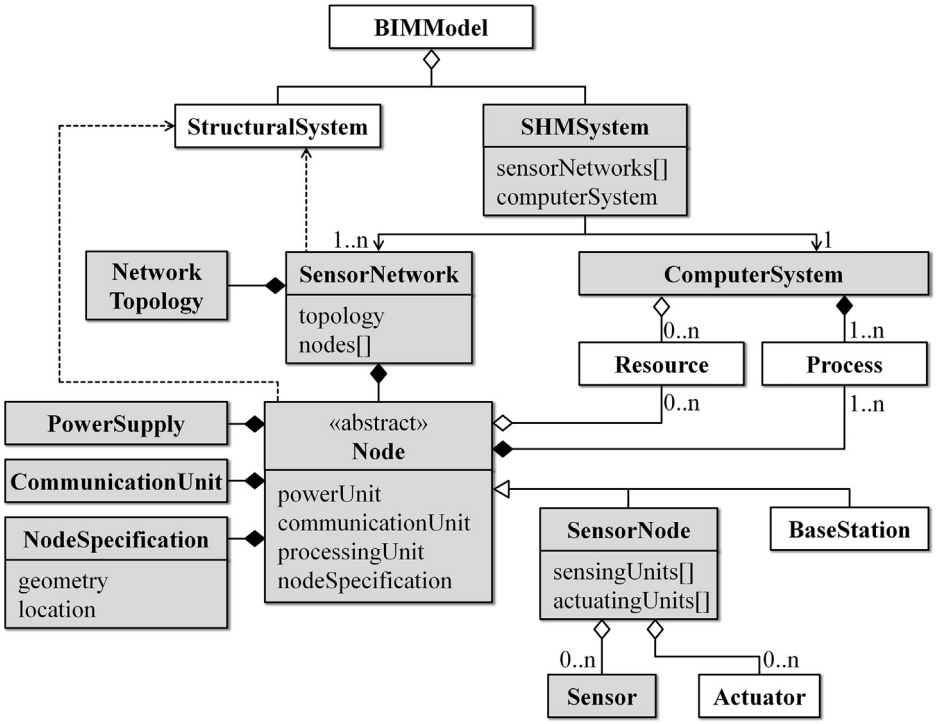


Fig. 2. Semantic model for SHM systems.

each node type includes processes, e.g. for on-board data analysis, as well as resources (such as flash memory for temporary data storage), which can optionally be assigned to nodes. Usually, at least one sensor or actuator is attached to a sensor node. However, since sensor nodes may be used solely for forwarding sensor data, sensors and actuators are considered optional components of sensor nodes.

In Fig. 2, the gray-colored subset of elements serves as a basis for illustratively demonstrating the IFC mapping of the semantic model as follows. The following sections focus on extending the IFC schema with monitoring-related information about

1. aggregating sensor node components into ***sensor nodes*** (such as sensors, actuators, communication units, or power supply units),
2. grouping sensor nodes into ***sensor networks***,
3. composing sensor networks and computer systems to ***SHM systems***,
4. associating ***network topologies*** to sensor networks, and
5. describing ***semantic relationships*** between components of the SHM system and components of the structural system being monitored.

4. Extending the IFC schema with semantic information on SHM systems

In this section, the IFC Monitor extension is presented. In total, three entities and one enumeration type are added to the IFC schema. The entities, gray-colored in Fig. 4, include sensor nodes (*IfcSensorNode*), sensor networks (*IfcSensorNetwork*), and SHM systems (*IfcSHMSystem*). The entities are integrated into the inheritance tree of the existing IFC schema according to the semantic meaning of each entity. In addition to the entities, an enumeration type (*IfcNetworkTopologyEnum*) is defined and added into the IFC Monitor extension for describing network topology types of sensor networks.

From a technical point of view, the IFC Monitor extension is defined

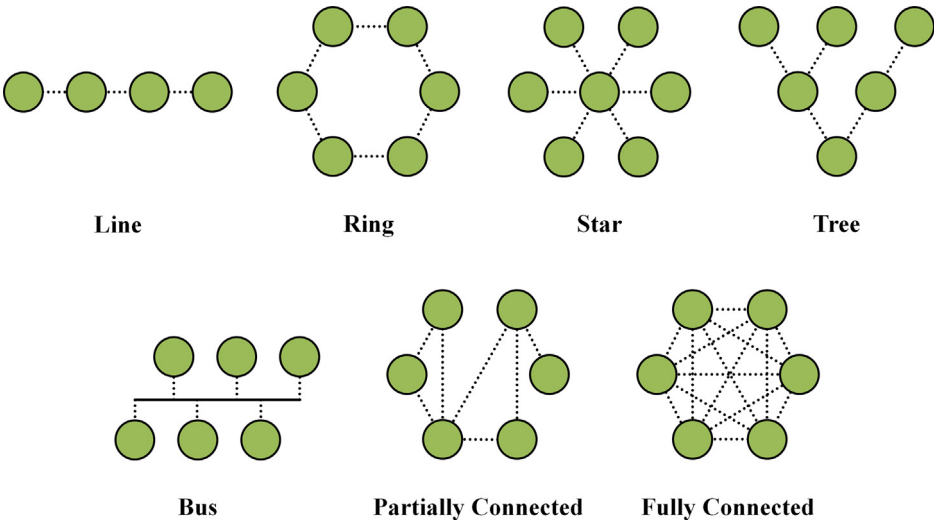


Fig. 3. Network topologies for SHM supported by the semantic model.

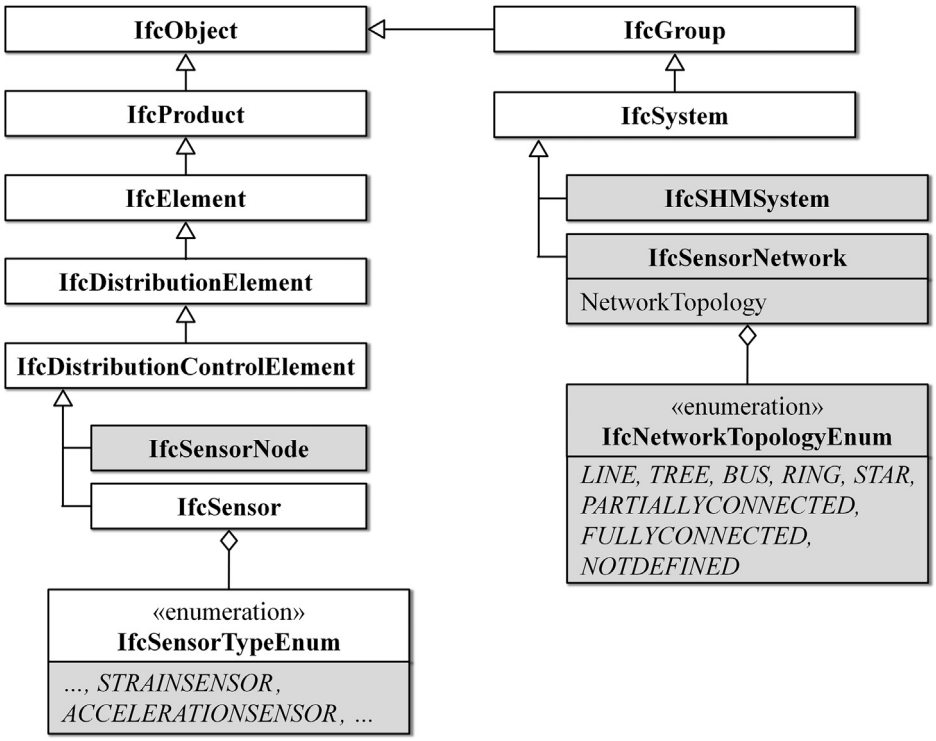


Fig. 4. Extract of the IFC Monitor extension with new entities, types, and enumerators gray-colored.

in compliance with the EXPRESS data modeling language, formalized in ISO 10303-21:2016 [37]. Illustrating new sensor types relevant to SHM, such as accelerometers and strain gauges, an extract of the IFC Monitor schema is shown in Listing 1. Also shown in Listing 1, the enumeration type *IfcNetworkTopologyEnum* for network topology descriptions is added to the IFC schema for characterizing sensor networks. Furthermore, the listing shows definitions of the entities *IfcSensorNode*, *IfcSensorNetwork* and *IfcSHMSystem* in association with super types. The definitions include specifications of entity-specific attributes (e.g. the attribute "NetworkTopology" of the entity *IfcSensorNetwork*) as well as entity-specific domain rules. Domain rules (also known as "where clauses") constrain the values of individual attributes. For the entities *IfcSensorNode*, *IfcSensorNetwork* and *IfcSHMSystem*, domain rules are introduced to restrict components, which may decompose or group the entities, to specific entity types. For example, *IfcSensorNetwork* entities can solely be grouped by *IfcSensorNode* entities.

Network topologies of sensor networks are informatively described with the enumeration type *IfcNetworkTopologyEnum*. In addition, connections between two components (e.g. between a sensor node and a computer system) are individually described using *IfcRelConnectsElements* relationship entities. Thus, information about network topologies is described redundantly; however, the enumeration type can be used to quickly retrieve information about network topologies of sensor networks without analyzing individual connections of the components. To avoid contradictions, the enumeration types must coincide with the individual connections. In the IFC Monitor extension, it is planned to add domain rules for that purpose.

A more detailed description of the entities *IfcSensorNode*, *IfcSensorNetwork*, and *IfcSHMSystem* is presented in the remainder of this section, followed by a verification of the IFC Monitor extension using test software of the official IFC certification program.

4.1. Sensor nodes

A sensor node, described by the *IfcSensorNode* entity (shown in Fig. 4 and Listing 1), is a subtype of the *IfcProduct* entity, because a sensor node is an object that appears in a spatial or geometric context, thus enabling assignments of geometric representations and locations to sensor nodes. Furthermore, sensor nodes extend the abstract *IfcElement* entity. Hence, sensor nodes inherit the capability to connect to other IFC entities, which can be used to logically describe network connections between sensor nodes within a sensor network. Semantically, a sensor node is an element of a distributed system (*IfcDistributionElement*), or, more precisely, a control element within a distribution system (*IfcDistributionControlElement*).

As introduced in the semantic model, every sensor node can be decomposed into several components, such as sensors and actuators, which are entities derived from the *IfcProduct* entity. Therefore, inherited from the abstract *IfcObjectDefinition* entity (shown in Fig. 1), decompositions of sensor nodes into components can be described using objectified relationship entities *IfcRelAggregates*. In particular, sensors described by *IfcSensor* entities, actuators described by *IfcActuator* entities, power supplies described by *IfcElectricFlowStorageDevice* entities, and communication units described by *IfcCommunicationsAppliance* entities compose single sensor node entities.

4.2. Sensor networks

A sensor network (*IfcSensorNetwork*), according to the semantic model, is a collection of nodes, i.e. sensor nodes and base stations. From a topological, non-geometrical grouping aspect, sensor networks describe distinct aggregations of nodes, described in the IFC schema through *IfcGroup* entities (Fig. 1). In the IFC Monitor extension, sensor networks exhibit characteristics of a system (*IfcSystem*), with a system representing an organized combination of interrelated components (here, the nodes), assembled for a common purpose to provide a certain service (here, the monitoring service). Relationships describing the grouping aspect between *IfcSensorNode* entities and *IfcSensorNetwork* entities are implemented by *IfcRelAssignsToGroup* relationship entities. Semantic relationships between sensor networks and spatial elements of the structural system are described by *IfcRelServicesBuildings* relationship entities. Network topology types are specified through the

```
TYPE IfcSensorTypeEnum =
  ENUMERATION OF (..., ACCELERATIONSENSOR, STRAINSENSOR, ...);
END_TYPE;

TYPE IfcNetworkTopologyEnum =
  ENUMERATION OF (LINE, TREE, BUS, RING, STAR, PARTIALLYCONNECTED, FULLYCONNECTED,
    NOTDEFINED);
END_TYPE;

ENTITY IfcSensorNode
  SUBTYPE OF (IfcDistributionControlElement);
  WHERE

    | NOT(SIZEOF(QUERY(Comp <* Rel.RelatedObjects | NOT(SIZEOF(['IFC4.IFCSENSOR',
    'IFC4.IFCACTUATOR', 'IFC4.IFCCOMMUNICATIONSAPPLIANCE',
    'IFC4.IFCELECTRICFLOWSTORAGEDEVICE'] * TYPEOF(Comp)) = 1))) = 0))) = 0;
END_ENTITY;

ENTITY IfcSensorNetwork
  SUBTYPE OF (IfcSystem);
    NetworkTopology: OPTIONAL IfcNetworkTopologyEnum;
  WHERE
   ApplicableComponents: SIZEOF(QUERY(Rel <* SELF\IfcGroup.IsGroupedBy |
    NOT(SIZEOF(QUERY(Comp <* Rel.RelatedObjects | NOT(SIZEOF(['IFC4.IFCSENSORNODE']
    * TYPEOF(Comp)) = 1))) = 0))) = 0;
END_ENTITY;

ENTITY IfcSHMSystem
  SUBTYPE OF (IfcSystem);
  WHERE
   ApplicableComponents: SIZEOF(QUERY(Rel <* SELF\IfcGroup.IsGroupedBy |
     NOT(SIZEOF(QUERY(Comp <* Rel.RelatedObjects | NOT(SIZEOF(
     ['IFC4.IFCSENSORNETWORK', 'IFC4.IFCCOMMUNICATIONSAPPLIANCE'] * TYPEOF(Comp))
      = 1))) = 0))) = 0;
END_ENTITY;
```

Listing 1. Schema file extract of the IFC Monitor extension.

enumeration type *IfcNetworkTopologyEnum* and are associated to a sensor network by the "NetworkTopology" attribute of the *IfcSensorNetwork* entity. Predefined values for *IfcNetworkTopologyEnum* describe different types of network topologies.

4.3. Structural health monitoring systems

A structural health monitoring system (*IfcSHMSystem*) is a composition of functionally related entities. Similar to *IfcSensorNetwork* entities, every SHM system is a subtype of *IfcSystem*. In the IFC Monitor extension, a SHM system is composed of sensor network(s) and a computer system. While computer systems are described by *IfcCommunicationsAppliance* entities (object type: COMPUTER), sensor networks are described by *IfcSensorNetwork* entities. The composition of sensor networks and computer systems to a SHM system is described by an *IfcRelAssignsToGroup* relationship entity.

4.4. Verification of the IFC Monitor extension

For verification purposes, test software of the official IFC certification program [32] is used to check the syntactic and semantic correctness of the IFC Monitor extension. For details on the official IFC certification program, the interested reader is referred to the website of the certification platform (*www.b-cert.com*). In the following paragraphs, an overview of the verification procedure conducted by the test software, implemented in Java programming language, is given, followed by the results achieved for the IFC Monitor extension. The verification procedure comprises three steps, syntactic checks, semantic checks, and unit testing.

i. **Syntactic checks.** A parser is designed for syntactic checks of EXPRESS-based IFC schemas to be verified. The IFC Monitor schema file is parsed with respect to the EXPRESS syntax specification to detect incorrect definitions of new, monitoring-related entities added to the IFC Monitor schema file that formally violate the syntax.
ii. **Semantic checks.** A compiler creates Java classes to provide semantic checks of the object model based on the schema to be verified [31]. If the parsing process of step (i) is completed without syntactic errors, a Java class for each entity specified in the IFC schema is generated. The Java classes reflect the inheritance tree of the entities specified in the IFC schema, and the classes provide methods to access the attributes of each entity. Furthermore, the Java classes provide test routines to check the semantic correctness of IFC models. As shown in Listing 2, the test routines are compiled to Java source code based on the definition of the domain rules specified in the IFC Monitor schema file. For efficient compilation, so-called "lambda expressions" [38], which can be nested several times, are used to implement the domain rule checks. Each domain rule that is checked within the test routines results in a Boolean value that must be "true" if the semantics are correct; otherwise, a semantic error is reported.
iii. **Unit tests.** Unit tests are devised to verify the IFC Monitor schema based on the Java classes created in step (ii). In unit tests, the test routines of the Java classes are tested according to whether the domain rules result in the expected Boolean values. Therefore, using

```
{
  ...
  public void whereRuleCheck(WhereRuleCheckResultList list)
  {
    super.whereRuleCheck(whereRuleCheckResultList);
    boolean ApplicableComponentsCheck = (BUILTINFUNCTIONS.SIZEOF(
      BUILTINFUNCTIONS.query(this.getIsGroupedBy_Inverse(),(Rel) ->
      !(BUILTINFUNCTIONS.SIZEOF(
      BUILTINFUNCTIONS.query(Rel.RelatedObjects,(Comp) -> !(BUILTINFUNCTIONS
      .SIZEOF(BUILTINFUNCTIONS.TYPEOF(Comp).retainAll(new STRING(
      "IFC4.IFCSENSORNETWORK",false),new STRING(
      "IFC4.IFCCOMMUNICATIONSAPPLIANCE",false)))==1.0)))==0.0)))==0.0);
    if( ! ApplicableComponentsCheck)
      reportError(list, "IfcSHMSystem:ApplicableComponents");
    ...
  }
```

Listing 2. Extract of the Java class created from the *IfcSHMSystem* entity.

```
// Create a new SHM system entity.
IfcSHMSystem ifcSHMSystem = new IfcSHMSystem();
// Create a relationship entity.
IfcRelAssignsToGroup ifcRelAssignsToGroup = new IfcRelAssignsToGroup();
ifcRelAssignsToGroup.setRelatingGroup(ifcSHMSystem);
SET<IfcObjectDefinition> RelatedObjects = new SET<>();
ifcRelAssignsToGroup.setRelatedObjects(RelatedObjects);
// Add components of SHM system to related objects.
RelatedObjects.add(new IfcSensorNetwork());
RelatedObjects.add(new IfcSensorNetwork());
RelatedObjects.add(new IfcCommunicationsAppliance());
// Now let's add a wall to related objects to violate the domain rule.
// Adding a wall must cause an error, since walls are no applicable components.
RelatedObjects.add(new IfcWall());
// Execute the domain rule check.
WhereRuleCheckResultList list = new WhereRuleCheckResultList();
ifcSHMSystem.whereRuleCheck(list);
// Print out the result.
//  >> OUTPUT: Domain rule IfcSHMSystem:ApplicableComponents violated!
System.out.println(list);
```

Listing 3. Unit test for the domain rule “ApplicableComponents” of the *IfcSHMSystem* entity.

the Java classes IFC test models are created, where domain rules are violated on purpose. By violating domain rules, it is tested whether the domain rules designed for the IFC Monitor extension work as expected. As an example, a unit test to check the domain rule “ApplicableComponents” of the *IfcSHMSystem* entity is shown in Listing 3. Therein, an IFC test model is created that includes a violation of a domain rule. As can be seen from Listing 3, a SHM system is created that is assigned to components of the SHM system or, more precisely, to two sensor networks, to a communication unit, and to a wall. Since a wall is no applicable component of a SHM system, the domain rule results, as expected, in a “false” indicating an incorrect state of the IFC test model.

A detailed overview of criteria for the verification procedure is shown in Table 1. As a result of the verification procedure, the IFC Monitor extension is positively verified, corroborating the compliance with the current IFC schema. Using the IFC Monitor extension, a prototype SHM system is designed for validation purposes, as described in the following section.

5. BIM-based modeling of a prototype SHM system using the IFC Monitor extension

To validate the BIM-based modeling approach for SHM systems, a prototype SHM system, which is installed on a laboratory test structure, is designed using the IFC Monitor extension. The validation aims at checking if the IFC Monitor schema is capable to precisely describe all monitoring-related information in compliance with the IFC standard. In the following subsections, the test structure and the prototype SHM system are presented. Then, BIM-based modeling of both the test structure and the prototype SHM system using the IFC Monitor extension is summarized in a concise overview, resulting in an IFC model of the test structure and of the prototype SHM system. Finally, the IFC model is validated using test software of the official IFC certification program.

5.1. Laboratory test structure and prototype SHM system

The prototype SHM system described with the IFC Monitor extension is installed on a four-story shear frame structure that serves as a test structure. As shown in Fig. 5, the structure consists of five rectangular 300 mm × 200 mm × 15 mm (length × width × thickness)

Table 1
Verification criteria and results.

Step	Criteria	Result
Syntactic checks	Formal compliance according to the grammar defined in ISO 10303-11 (EXPRESS); in particular, compliance to ● notational conventions according to the language specification syntax, which is defined by productions, expressions, terms, factors, identifiers, literals, groups, options and repetitions, ● defined sets of characters for digits, letters, special characters, whitespaces, etc., ● reserved keywords, and ● sequence of elements of the language specification syntax.	✓
	EXPRESS-compliant declaration of ● entities (“entity data types”) and types (“defined data types”), ● enumerations (“enumeration data types”) and selections (“select data types”), ● subtype and supertype constraints, ● attributes of entities (“explicit”, “derived”, and “inverse”), ● domain rules (“where clauses”) and uniqueness rules, and ● functions, procedures, and expressions.	✓
Semantic checks	Compilation of the IFC Monitor schema into Java classes preparing semantic checks with the extended schema; in particular, the compiler generates ● Java classes for each entity, type, and enumeration considering the inheritance hierarchy specified and ● Java interfaces for each selection declaring ● accessor and mutator methods for explicit attributes, ● accessor methods for derived and inverse attributes, and ● methods to semantically check domain rules and uniqueness rules of models	✓
Unit testing	Conformity with expected results of domain rules for both correct and incorrect states of models.	✓



Fig. 5. Physical implementation of the prototype SHM system on a laboratory test structure.

aluminum slabs fixed to four aluminum columns of 20 mm × 2 mm rectangular cross sections. The five slabs form four stories with a height of 300 mm each.

The prototype SHM system features four wireless sensor nodes, a base station, and a computer system. The wireless sensor nodes are connected in star topology. As shown in Fig. 5, one sensor node is placed at the center of each story, while the base station serves as a gateway between the sensor nodes and the computer system. The monitoring objective of the prototype SHM system is to perform automated damage detection by extracting the mode shapes of the test structure through frequency domain decomposition [39]. Therefore, local data is processed and information is exchanged among the sensor nodes. Essentially, local data processing comprises collecting acceleration response measurements and transforming the collected measurements into the frequency domain via a fast Fourier transform algorithm embedded into the sensor nodes. Details on the embedded damage detection approach can be found in [40,41].

5.2. IFC-compliant modeling of the laboratory test structure and the prototype SHM system

The laboratory test structure (i.e. the structural system to be monitored) and the prototype SHM system are modeled with the IFC Monitor extension on a coherent BIM basis. While the test structure is conventionally modeled using commercial BIM software (Fig. 6), modeling monitoring-related information, which is not included in commercial BIM software, is done with the same test software that is used for the verification of the IFC Monitor schema. Using the IFC model of the test structure as a basis, monitoring-related information that describes the prototype SHM system is added to the IFC model regarding (i) the semantic composition of the SHM system, (ii) the network topology of the system, and (iii) the semantic relationships between components of the SHM system and components of the structural system to be monitored.

For describing and exchanging IFC models, the standard for the exchange of product model data (STEP), formalized in ISO-10303-21:2016 [42], is a widely used, well-established format. Listing 4 shows an extract of the STEP-based description of the IFC model describing the laboratory test structure and the prototype SHM system. For the sake of clarity, attributes compulsory in the IFC schema, such as the global identifiers of entities, are omitted in Listing 4. A schematic representation of the IFC model is shown in Fig. 7.

In the IFC model shown in Listing 4, the prototype SHM system (#1) is composed of a computer system (#2) and a sensor network (#3). Grouping relationships between SHM system and its components, i.e. the computer system and the sensor network, are realized using a relationship entity *IfcRelAssignsToGroup* (#30). The network topology of the sensor network, a star topology, is defined by the last parameter in the parameter list of the *IfcSensorNetwork* entity (#3).

The sensor network is composed of four sensor nodes (#5, #6, #7, #8) and a base station (#4) that are associated with the sensor network through an *IfcRelAssignsToGroup* relationship entity (#31). For illustration purposes, the base station is modeled by an *IfcSensorNode* entity. With *IfcRelConnectsElements* relationship entities, such as entity #35 connecting base station #4 with sensor node #5, explicit logical



Fig. 6. BIM-based model of the laboratory test structure and of the prototype SHM system.

connections between two nodes are described. The explicit logical connections between the nodes of the sensor network must be consistent with the informal description of the network topology of the sensor network. Relationships between sensor networks and components of the structural system is illustratively described using an *IfcRelServicesBuildings* relationship entity (#33) that specifies the relationship between the sensor network (#3) and a story of the laboratory test structure (#20).

The decomposition of sensor nodes into components is implemented using an *IfcRelAggregates* relationship entity; here, sensor node #6 contains four components, an acceleration sensor (#9), a temperature sensor (#10), a power supply unit in form of a battery (#11), and a communication unit in form of an antenna (#12). For demonstration purposes, sensor node #6 is semantically referenced by a structural component (#21) using an *IfcRelAssignsToProduct* relationship entity (#34).

5.3. Validation of the IFC model

In a final step, the STEP-based exchange file of the IFC model describing the laboratory test structure and the prototype SHM system is validated using the test software previously introduced. The test software, adapted to the IFC Monitor extension, contains test algorithms to validate (i) if the exchange file is free of syntactic errors and (ii) if the IFC model meets the semantic requirements of a well-defined IFC model as specified in the official IFC certification program. In addition, the domain rules specified in the IFC Monitor extension are checked for compliance with the semantic model for SHM systems introduced in this study. The validation aims to verify IFC-compliant descriptions of SHM systems based on the IFC Monitor extension. Therefore, the IFC model is checked with respect to several criteria, e.g., covering aspects of information that are missing or of information that are incorrectly modeled. A detailed overview of criteria for the validation is provided in Table 2. As a result, the IFC model of the laboratory test structure and the prototype SHM system meets all requirements. It is concluded that the IFC Monitor extension proposed in this study provides well-defined descriptions of SHM systems in compliance with the IFC standard.

6. Summary and conclusions

In this study, the IFC schema has been extended to describe

```
#2= IFCCOMMUNICATIONSAPPLIANCE($,$,'Computer System 01',$,$,$,$,$,.COMPUTER.);
#3= IFCSENSORNETWORK($,$,'Sensor Network 01',$,$,.STAR.);
#4= IFCSENSORNODE($,$,'Base Station 01',$,$,$,$,$);
#5= IFCSENSORNODE($,$,'Sensor Node 01',$,$,$,$,$);
#6= IFCSENSORNODE($,$,'Sensor Node 02',$,$,$,$,$);
#7= IFCSENSORNODE($,$,'Sensor Node 03',$,$,$,$,$);
#8= IFCSENSORNODE($,$,'Sensor Node 04',$,$,$,$,$);
#9= IFCSENSOR($,$,'Sensor 01',$,$,$,$,$,.ACCELERATIONSENSOR.);
#10= IFCSENSOR($,$,'Sensor 02',$,$,$,$,$,.TEMPERATURESENSOR.);
#11= IFCELECTRICFLOWSTORAGEDEVICE($,$,'Power Unit 01',$,$,$,$,$,.BATTERY.);
#12= IFCCOMMUNICATIONSAPPLIANCE($,$,'Communication Unit 01',$,$,$,$,$,.ANTENNA.);
...
#20=IFCBUILDINGSTOREY($,$,'Ground Floor',$,$,$,$,$,$,$);
#21=IFCSLAB($,$,'Slab 01',$,$,$,$,$,$,$);
...
#30= IFCRELASSIGNSTOGROUP($,$,'Grouping SHM System 01',$,(#2,#3),$,#1);
#31= IFCRELASSIGNSTOGROUP($,$,'Grouping Sensor Network 01',$,(#4,#5,#6,#7,#8),$,#3);
#32= IFCRELAGGREGATES($,$,'Aggregation Sensor Node 02',$,#6,(#9,#10,#11,#12));
#33= IFCRELSERVICESBUILDINGS($,$,'Service NW01-GroundFloor',$,#3,(#20));
#34= IFCRELASSIGNSTOPRODUCT($,$,'Assign SN02-Slab01',$,(#21),$,#6);
#35= IFCRELCONNECTSELEMENTS($,$,'Connection Base-SN04',$,$,#4,#8);
#36= IFCRELCONNECTSELEMENTS($,$,'Connection Base-Computer01',$,$,#2,#4);
```

Listing 4. Extract of the STEP-based description of the prototype SHM system.

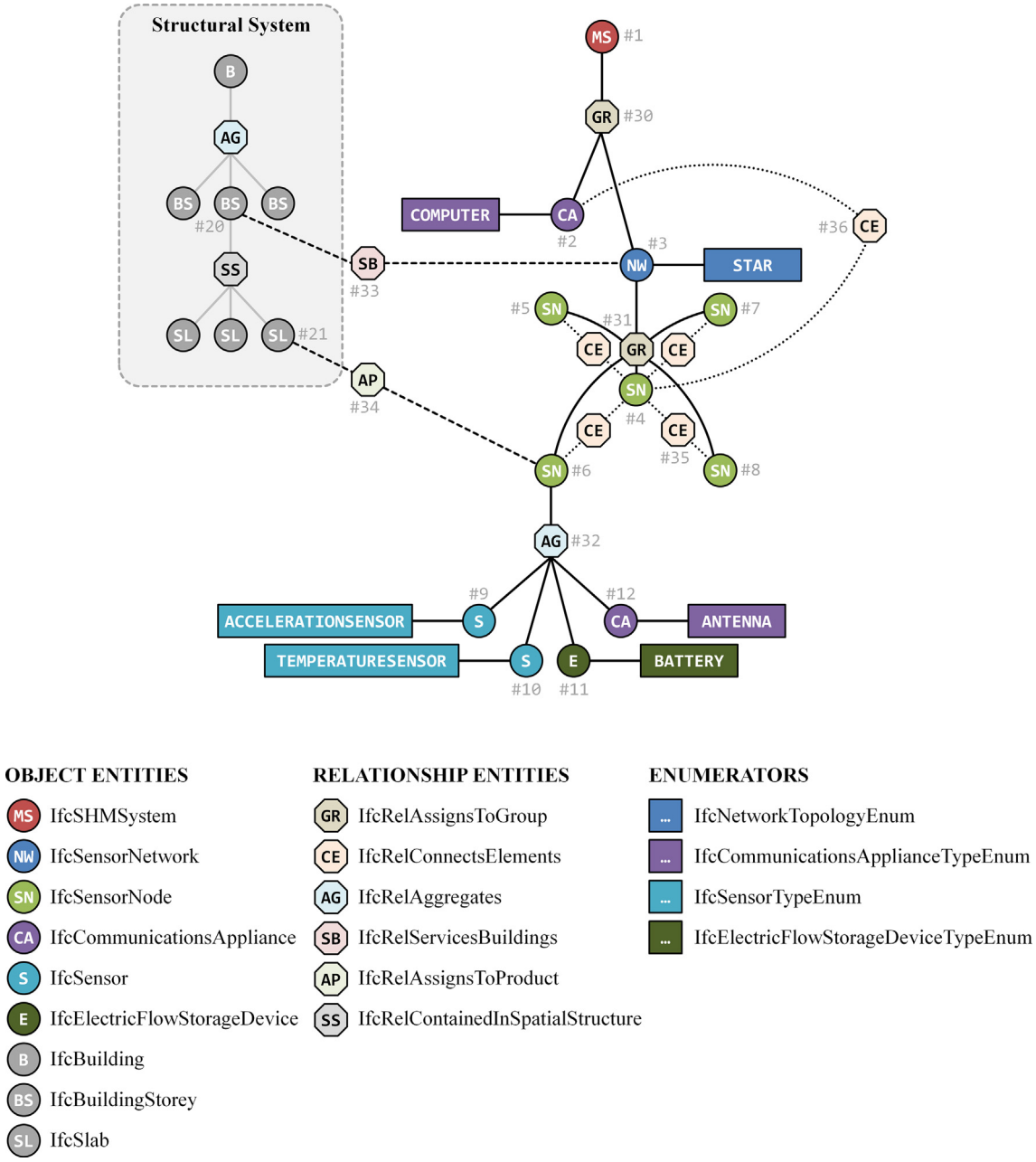


Fig. 7. Extract of the IFC model of the laboratory test structure and of the prototype SHM system as a schematic visualization.

monitoring-related information for BIM-based modeling of SHM systems. A semantic model to formally describe SHM systems has been proposed, which serves as a technology-independent metamodel for describing monitoring-related information. Based on the semantic model, a concept to describe monitoring-related information using the IFC schema has been developed. Therefore, the IFC schema has been extended to provide a well-defined description of monitoring-related information, referred to as "IFC Monitor" extension. For validating the BIM-based modeling approach, a prototype SHM system, in association with a laboratory test structure to be monitored, has been modeled based on the IFC Monitor extension. For verification, both the IFC Monitor extension and the IFC model of the prototype SHM system have been checked with test software of the official IFC certification program [31,32]. The test software provides a framework, e.g., for parsing EXPRESS-based schemas (such as the IFC schema), for parsing STEP-based exchange files (such as IFC models), and for syntactically and semantically checking of EXPRESS-based schemas and of STEP-based exchange files.

The results show that the IFC Monitor extension facilitates the description of monitoring-related information, precisely information about (i) semantic compositions of SHM systems, (ii) network topologies, and (iii) semantic relationships between components of the SHM systems and components of structural systems being monitored. It can be concluded that with the proposed IFC Monitor extension, monitoring-related information can correctly and precisely be described on a well-defined, formal basis in compliance with the IFC schema, supporting the design process, the documentation, and the information exchange of SHM systems.

In future work, enhancements of the IFC Monitor extension could focus on the description of communication-related information, such as communication protocols. Representing another open research issue relevant to formally describing SHM systems, the system dynamics inherent to SHM systems cannot yet adequately be described with existing IFC-based modeling approaches and could be addressed in future

Table 2
Validation criteria and results.

Step	Criteria	Result
Syntactic checks	Formal compliance according to the grammar defined in ISO 10303-21 (STEP); in particular, compliance to • notational and typographical conventions according to the specification of the exchange structure syntax, • defined sets of characters for digits, letters, special characters, whitespaces, etc., • data type encodings and clear text encodings, and • reserved keywords.	✓
	Conformance of header section with the header section schema.	✓
	Conformance of data section with the IFC Monitor schema including • use of declared data types only and • number and type of attributes specified.	✓
Semantic checks	Accordance of header section to IFC header definition.	✓
	Presence of non-optional attributes.	✓
	Accordance to geometry restrictions specified in the IFC documentation.	✓
	Conformance with implementers' agreements of the IFC standard.	✓
	Compliance with domain rules and uniqueness rules.	✓

research efforts.

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